

VAM Split Strategy

Executive Summary

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What Is the VAM Split Strategy?

The VAM Split Strategy is a systematic, rules-based trading system designed to outperform the S&P 500 over full market cycles. It uses three complementary sub-strategies that operate inside a single capital pool. No capital is ever double-counted, and the system is hard-capped at 99% invested at all times.

The Three Sub-Strategies

Strategy	Name	What It Does	When Active
Strategy 1	VAM Bull Engine	Holds UPRO (75%) + TQQQ (25%) in bull markets. Trims exposure during volatility. Goes to cash on crisis signals.	Always (primary engine)
Strategy 2	Predatory Short (SPXU)	Deploys 50% of capital into SPXU (3x inverse S&P;) to profit from continued market decline. Crash insurance.	Only when Strategy 1 is in CASH
Strategy 3	Safety Valve (SVIX)	Deploys 10–30% into SVIX (short volatility) to profit as panic subsides after a crash. Available from March 2022.	Only when Strategy 1 is in CASH

ETF Reference

ETF	3x Factor	Tracks	Role in Strategy
UPRO	3×	S&P; 500 (SPY)	Primary bull engine — largest holding
TQQQ	3×	NASDAQ-100 (QQQ)	Growth sleeve — adds tech exposure
SPXU	3× Inverse	S&P; 500 (short)	Crash insurance — profits in bear markets
SVIX	1×	Short VIX futures	Volatility harvester — profits as panic fades
SPY	1×	S&P; 500	Benchmark comparison

Backtest Parameters

Parameter	Value
Data Source	DataBento (XNAS.ITCH) production daily OHLCV bars
Backtest Period	2019-02-15 to 2025-12-30 (6.9 years)
Initial Capital	\$100,000
Execution Model	T+1 (signal at close → execute at next-day open)
Slippage	5 bps normal / 20 bps on stress days (VIX ≥ 25)

Parameter	Value
Commission	IBKR percentage-based model (~0.35% of trade value, min \$1)
Risk-Free Rate	4.0% annual (subtracted before Sharpe / Sortino calculation)
Allocation Cap	99% max invested at any time

Step-by-Step Results

Each step represents a backtest run with progressively more sophisticated logic. Steps 3 and 4 show standalone sub-strategy performance (active only during CASH periods).

Step	Strategy	Final Value	CAGR	Sharpe	Max DD	Trades
Step 1	UPRO Only (4-state)	\$333,662	+19.2%	0.594	−38.5%	90
Step 2	UPRO + TQQQ (7-state)	\$408,259	+22.7%	0.661	−49.9%	294
Step 3	SPXU Predatory Short*	\$69,476	−5.2%	-0.381	−48.3%	48
Step 4a	SVIX Safety Valve*	\$109,808	+1.2%	-1.735	−2.7%	6
Step 4 Combined	Full Portfolio (All 3 strategies)	\$296,800	+17.1%	0.503	−50.1%	54
SPY Benchmark	Buy & Hold S&P 500	—	+14.1%	—	—	—

* Steps 3 and 4a show standalone sub-strategy performance. They activate only during Step 2 CASH periods and are designed for crash protection, not standalone alpha. Standalone returns are expected to be low or negative.

Key Takeaways

- Strategy 1 (Step 2) delivered +22.7% CAGR vs SPY's +14.1% — alpha of +8.6% annualised.
- Strategy 1 hit a maximum drawdown of −49.9% during the 2022–2023 bear market. SPY's max drawdown over the same period was approximately −25%. The leveraged exposure amplifies both gains and losses.
- The combined portfolio (Step 4) shows +17.1% CAGR and Sharpe 0.503 — lower raw return than Step 2 alone but with crash protection active during CASH periods.
- SVIX has a very short live history (launched March 2022). Its standalone metrics cover only 3 full-cycle cash periods.

Portfolio Performance — Annual Returns

Annual returns for Strategy 1 (UPRO only) and Strategy 1+2 (UPRO + TQQQ). SPY CAGR of 14.07% shown for reference. Year-by-year SPY breakdown not available from DataBento output.

Year	2019	2020	2021	2022	2023	2024	2025	CAGR
Step 1 (UPRO)	+14.7%	+20.4%	+60.2%	−29.1%	+28.4%	+27.7%	+32.1%	+19.2%
Step 2 (UPRO+TQQQ)	+31.0%	+29.3%	+52.1%	−39.5%	+39.9%	+50.9%	+26.4%	+22.7%
SPY (Buy & Hold)	—	—	—	—	—	—	—	+14.1%

Drawdown Summary

Metric	Step 1 (UPRO)	Step 2 (UPRO+TQQQ)	Step 4 Combined
Maximum Drawdown	−38.5%	−49.9%	−50.1%
Drawdown Date	2023-03-10	2023-03-22	2023-03-22
Calmar Ratio	0.498	0.456	0.342
Sharpe Ratio	0.594	0.661	0.503
Sortino Ratio	0.731	0.823	0.602

State Distribution (Step 2)

State	Description	% of Trading Days
BULL_100	Fully invested (74.25% UPRO + 24.75% TQQQ)	56.8%
CASH	Kill switch active — $VIX \geq 30$ or SPY below 200-SMA	21.9%
DEFENSIVE_BOT H	Both SPY and QQQ below respective 50-SMAs	7.5%
BULL_TRIMMED	RSI trim active ($RSI \geq 75$)	4.9%
DEFENSIVE_QQQ	QQQ below 50-SMA; SPY still bullish	3.8%
SMA_RECOVERY	Re-entering bull from CASH (intermediate state)	3.4%
DEFENSIVE_SPY	SPY below 50-SMA; QQQ still bullish	1.8%

Risk Disclosures

1. Leveraged ETF Risk

UPRO, TQQQ, and SPXU are 3× leveraged ETFs that reset daily. They are subject to volatility decay in sideways or choppy markets. A prolonged sideways market can erode value even without a directional move. These instruments are not designed for buy-and-hold.

2. Backtest Period Limitation

This backtest covers 6.9 years (2019-02-15 to 2025-12-30). It does NOT include the 2000–2003 dot-com crash or the 2007–2009 financial crisis. A leveraged ETF strategy during either of those periods would have faced extreme drawdowns. Past performance in 2019–2025 does not predict performance in future multi-year bear markets.

3. Maximum Drawdown

Strategy 1 experienced a –49.9% drawdown at peak (Step 2). On a \$100,000 initial capital base, this represents a drop to approximately \$50,000 at the worst point. The kill switch limited further losses but did not eliminate drawdown during the 2022 rate-hike cycle.

4. SVIX Short History

SVIX (1× Short VIX Futures ETF) launched on March 30, 2022. It has fewer than 4 years of live trading history. The 2022 VIX spike occurred shortly after its launch. Its behaviour during a prolonged volatility regime (2008–2009 equivalent) cannot be backtested with live data.

5. 2022 Underperformance

In 2022, Strategy 2 (UPRO+TQQQ) lost –39.5%. SPY lost approximately –18.2% the same year. The kill switch prevented a larger loss (UPRO unhedged fell –65% in 2022), but the strategy still underperformed a simple SPY position during the crash year.

6. Backtest ≠ Future Results

All results shown are historical simulations. They assume perfect execution at next-day open prices with modelled slippage. Real-world execution may differ due to liquidity, market impact, and regime changes. Past backtest results are not a guarantee of future performance.